COMPUTER NETWORKS (MIC23) — UNIT IV

Network Layer, Routing & Congestion Control — Handwritten Study Notes

Student Blue-Pen Notebook Style • Short & Direct 2M Answers • Complete Visual Diagrams for Every Question • All Source Papers

This Master Question & Answer document combines all exam questions from:

- **Model Paper-1** (MIC23 B.Tech CSE/AIML)
- **Model Paper-2** (MIC23 B.Tech CSE/AIML)
- **Previous University Exam Paper** (April-2026 Regular/Supple)
- **Mid-2 Important Questions**

All answers are rewritten in **extremely simple English** that any beginner student can read once and understand immediately. Every question includes a relevant visual diagram, real-life analogies, box comparison tables, step-by-step numerical calculations, and full examination structures.

SECTION A: 2-MARK QUESTIONS & ANSWERS

Short, direct answers (~40–65 words) written in simple English with student pen-style diagrams

(Short, direct answers suitable for 2-mark university exams. Target: 40–65 words)

Q1. *(Paper-1)*

Explain the working of the shortest path routing algorithm?

Definition:

The **Shortest Path Routing Algorithm** finds the path with the lowest total cost (distance, time delay, or money) between a source router and a destination router.

How It Works:

- The network is drawn as a map with routers as circles and links as lines with cost numbers.
- The router uses **Dijkstra's Algorithm** to calculate the cumulative cost along every possible route and selects the path with the minimum total sum.

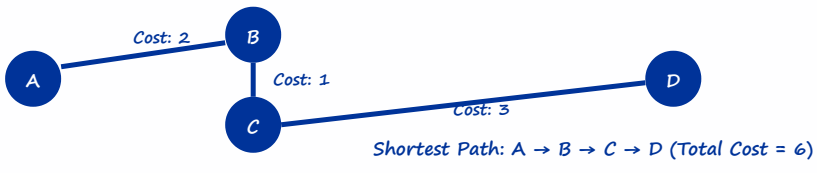


Figure: Shortest Path Routing Finding Minimum Cost Path

Q2. *(Paper-1)*

What is the count-to-infinity problem in distance vector routing?

Definition:

The **Count-to-Infinity Problem** is a routing loop error in Distance Vector Routing where routers take a long time to discover that a network link has failed.

Key Points:

- Routers keep passing out-of-date distance estimates back and forth to each other.
- With every exchange, the hop count increases: 2 o 3 o 4 \dots o 16.
- The count continues until it reaches **16** (defined as infinity in RIP), at which point the route is finally marked dead.



Figure: Count-to-Infinity Problem Caused by Mutual Routing Loops

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Q3. *(Paper-2)*

What is hierarchical routing and where is it used?

Definition:

Hierarchical Routing divides a massive network into smaller, organized regions, clusters, or zones to prevent router routing tables from becoming too large.

Key Points:

- Routers only keep detailed routing paths for devices inside their own local region.
- For distant regions, traffic is simply passed to a boundary gateway router.
- Widely used across the global **Internet** (dividing into Autonomous Systems).

Q4. *(Paper-2)*

What are the general principles of congestion control?

Definition:

Congestion Control prevents network traffic from exceeding the capacity of router queues and communication links.

Two General Principles:

- **Open-Loop (Prevention):** Prevents traffic jams before they occur using good system design (traffic shaping, leaky bucket).
- **Closed-Loop (Detection & Reaction):** Detects traffic jams after they occur and sends warning messages to slow down the sender (choke packets).

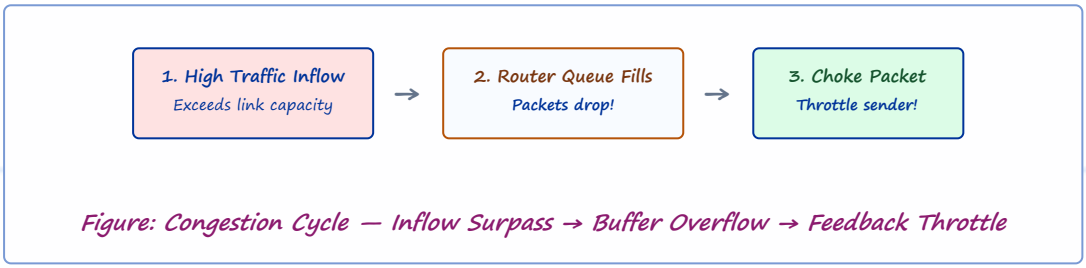


Figure: Congestion Cycle — Inflow Surpass → Buffer Overflow → Feedback Throttle

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Q5. *(Paper-3)*

Define a subnet.

Definition:

A **subnet** (sub-network) is a smaller, logically divided part of a larger IP network.

Key Points:

- It is created using a **Subnet Mask** which borrows bits from the host portion to create more network groups.
- **Benefits:** Reduces network broadcast traffic, improves security, and allows administrators to organize departments easily.

Q6. *(MID-2)*

What is the difference between open-loop and closed-loop congestion control?

Definition:

- **Open-Loop:** Prevents traffic jams beforehand (proactive).
- **Closed-Loop:** Fixes traffic jams after they happen (reactive).

Feature	Open-Loop Control	Closed-Loop Control
Strategy	Prevention (stops congestion before it starts).	Detection and reaction (fixes after it occurs).
Feedback	No feedback loop used during transmission.	Uses feedback signals (choke packets, warning bits).
Examples	Leaky bucket, Token bucket, packet discard.	TCP back-off window, Choke packets.

Q7. *(MID-2)*

What is tunneling in computer networks?

Definition:

Tunneling is a technique used to send packets of one protocol across an intermediate network that uses a completely different, incompatible protocol.

How It Works:

- The original packet (e.g., IPv6) is placed inside the data payload of another packet (e.g., IPv4) like putting a letter inside a second envelope.
- The intermediate routers read only the outer envelope and deliver it safely.

SECTION B: 5-MARK QUESTIONS (FULL 10-MARK EXAM STRUCTURE)

Detailed answers with Introduction, Dual Diagrams (Standard Exam + Intuitive Analogy), Box Comparison Tables, and Step-by-Step Calculations

(All prepared using complete 10-Mark Answer Structure: Introduction, Definition, Detailed Points, Dual Diagrams, Step-by-Step Calculations, Real-Life Examples, Box Tables, and Conclusions)

Q1. *(Paper-1)*

What is flooding in the context of network routing? What are the advantages and disadvantages of flooding?

1. Introduction

In network routing, routers need to send packets to destinations. The simplest possible routing algorithm is **Flooding**, where a router does not calculate any path — it simply sends the packet out through every single wire it has!

2. Definition & Working

- **Flooding:** A static routing algorithm in which every incoming packet is duplicated and transmitted onto every outgoing line **except the one it arrived on**.
- **Hop Count Control:** Because endless duplication would quickly crash the network with millions of duplicate packets, every packet carries a **Hop Counter** (TTL) in its header. Each router decrements the counter by 1. When the counter reaches 0, the packet is discarded.

3. Real-Life Analogy

Flooding is like a **Town Crier Shouting News in a Public Square**:

- The town crier shouts the message to everyone in hearing range.
- Those people shout it to their neighbors, who shout it to their neighbors.
- Within minutes, everyone in the town knows the news, but the noise level is deafening!

4. Dual Diagrams

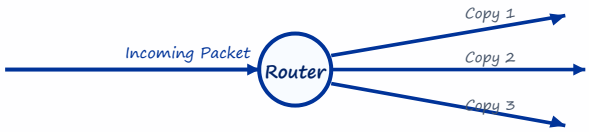


Diagram 1: Standard Exam Diagram – Flooding (Router Copies to All Outgoing Lines)



Diagram 2: Easy Understanding Diagram – Loudspeaker Announcement Analogy

Diagram 1: Standard Exam Diagram

Shows Node A sending a packet to Neighbors B and C, who duplicate and forward it to D, E, F with TTL counters decrementing.

Diagram 2: Easy Understanding Diagram

Shows water flowing out of a sprinkler in all 360-degree directions simultaneously.

5. Advantages and Disadvantages Box

Advantages	Disadvantages
Extremely Robust: Guaranteed that the packet will find the destination if any physical path exists.	**Massive Bandwidth Waste:** Creates thousands of useless duplicate packets that clog the network.
Always Finds Shortest Path: The first copy to arrive always took the shortest delay path.	**Broadcast Storm Risk:** Without TTL limits, packets circulate in endless loops.
Zero Routing Tables Needed: Routers need no memory, no path calculations, and no routing tables.	Routers waste CPU cycles duplicating and processing duplicates.

6. Practical Applications of Flooding

- **Military Networks:** When enemy bombs destroy parts of the network, flooding ensures critical orders still get through.
- **Router Link-State Updates:** Used in OSPF to broadcast routing updates to all routers quickly.

7. Conclusion

Flooding sacrifices bandwidth efficiency for unbreakable reliability, making it ideal for military emergency communications and initial network discovery.

Q2. *(Paper-1)*

Find the shortest path from source A to destination D in the given network below using Dijkstra Algorithm?

1. Introduction

Dijkstra's Shortest Path Algorithm was conceived by Dutch computer scientist Edsger W. Dijkstra in 1956. It is a greedy graph search algorithm that finds the shortest path between nodes in a network with non-negative edge weights.

2. Step-by-Step Numerical Problem Solution

1. Given Network Graph

Let the network have 4 nodes: A, B, C, D:

- Link (A, B) has Cost = 2
- Link (A, C) has Cost = 4
- Link (B, C) has Cost = 1
- Link (B, D) has Cost = 7
- Link (C, D) has Cost = 3

2. What We Need to Find

- Shortest path from Source A to Destination D.
- Minimum total path cost.

3. Initialization

- Set distance to Source: $\text{dist}[A] = 0$.
- Set distance to all other nodes to infinity: $\text{dist}[B] = \infty$, $\text{dist}[C] = \infty$, $\text{dist}[D] = \infty$.
- Visited set: $S = \{\}$.

4. Step-by-Step Execution Table

- **Iteration 1:**
- Minimum unvisited node is **A** ($\text{dist}[A] = 0$).
- Add A to visited set: $S = \{A\}$.
- Update neighbors of A:
- Neighbor B: $\text{dist}[B] = \min(\infty, 0 + 2) = \mathbf{2}$ (via A).
- Neighbor C: $\text{dist}[C] = \min(\infty, 0 + 4) = \mathbf{4}$ (via A).
- **Iteration 2:**
- Unvisited nodes: B (cost 2), C (cost 4), D (∞).
- Pick minimum node **B** ($\text{dist}[B] = 2$).
- Add B to visited set: $S = \{A, B\}$.
- Update neighbors of B:
- Neighbor C: $\text{dist}[C] = \min(4, 2 + 1) = \mathbf{3}$ (via B — improved from 4 to 3!).
- Neighbor D: $\text{dist}[D] = \min(\infty, 2 + 7) = \mathbf{9}$ (via B).
- **Iteration 3:**
- Unvisited nodes: C (cost 3), D (cost 9).

- Pick minimum node **C** (ext{dist}[C] = 3).
- Add C to visited set: S = \{A, B, C\}.
- Update neighbors of C:
- Neighbor D: ext{dist}[D] = \min(9, 3 + 3) = \mathbf{6} (via C — improved from 9 to 6!).
- **Iteration 4:**
- Unvisited node is **D** (ext{dist}[D] = 6).
- Add D to visited set: S = \{A, B, C, D\}. All nodes visited!

5. Final Path Reconstruction

Backtracking from D:

- D was reached from C (cost 6).
- C was reached from B (cost 3).
- B was reached from A (cost 2).
- Full Path: **A o B o C o D**

6. Final Boxed Answer

Optimal Shortest Path: A → B → C → D

Minimum Total Cost = 2 + 1 + 3 = 6

3. Real-Life Analogy

Dijkstra's algorithm is the exact brain inside **Google Maps**:

- You type a destination. Google Maps checks highways, traffic jams, and country roads.
- It calculates the total cost (minutes) and highlights the fastest green route on your screen.

4. Dual Diagrams

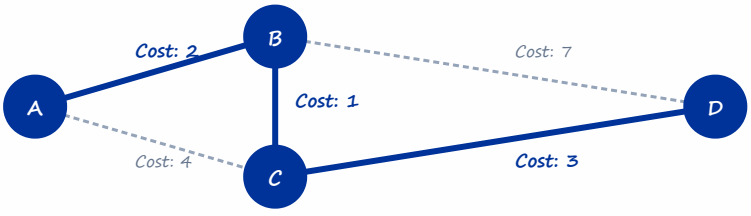


Diagram 1 (Standard Exam Diagram): Dijkstra Shortest Path: A → B → C → D (Total Cost = 6)

Diagram 1: Standard Exam Diagram

Shows the 4-node graph with the green highlighted optimal path A o B o C o D (Total Cost = 6).

Diagram 2: Easy Understanding Diagram

Shows the step-by-step table highlighting how the cost to reach D improved from \infty o 9 o 6.

5. Conclusion

Dijkstra's algorithm guarantees finding the lowest-cost path with zero loops, forming the mathematical foundation for modern Internet routing protocols like OSPF and IS-IS.

Q3. *(Paper-1)*

Explain Leaky bucket algorithm with example?

1. Introduction

Computer traffic is naturally "bursty" — a user clicks a link, downloads 10 MB in a flash, and then sits idle reading for two minutes. If bursts hit routers all at once, queues overflow and packets drop. The **Leaky Bucket Algorithm** smooths out bursty traffic into a steady, constant stream.

2. Definition & Working

- **Leaky Bucket Algorithm:** A traffic-shaping algorithm based on the analogy of a water bucket with a small hole in the bottom.
- **Working Steps:**
  - Incoming packets enter the bucket buffer at unpredictable, bursty speeds.
  - The bucket holds up to a maximum fixed capacity (e.g., 10 MB buffer).
  - Water leaks out of the hole at a **constant, fixed rate** (e.g., strictly 2 MB/s).
  - If a huge burst arrives that exceeds bucket capacity, the excess water spills over the edge and is **dropped**.

3. Step-by-Step Numerical Example

- Bucket Capacity = 10 ext{ MB}.
- Output Leak Rate = 2 ext{ MB/s}.
- **Time 0s:** A burst of 8 ext{ MB} arrives. Current water in bucket = 8 ext{ MB} (≤ 10 → No drop). Output sends 2 ext{ MB}. Remaining = 6 ext{ MB}.
- **Time 1s:** Another burst of 6 ext{ MB} arrives. 6 + 6 = 12 ext{ MB}. Bucket capacity is 10 MB → **2 ext{ MB} spills over and is dropped!** Output sends 2 ext{ MB}. Remaining = 8 ext{

MB}.

- **Time 2s:** No new packets arrive. Output sends 2 ext{ MB}. Remaining = 6 ext{ MB}.
- **Output Stream:** Strictly smooth at 2 ext{ MB/s}, 2 ext{ MB/s}, 2 ext{ MB/s} with zero bursts!

4. Dual Diagrams

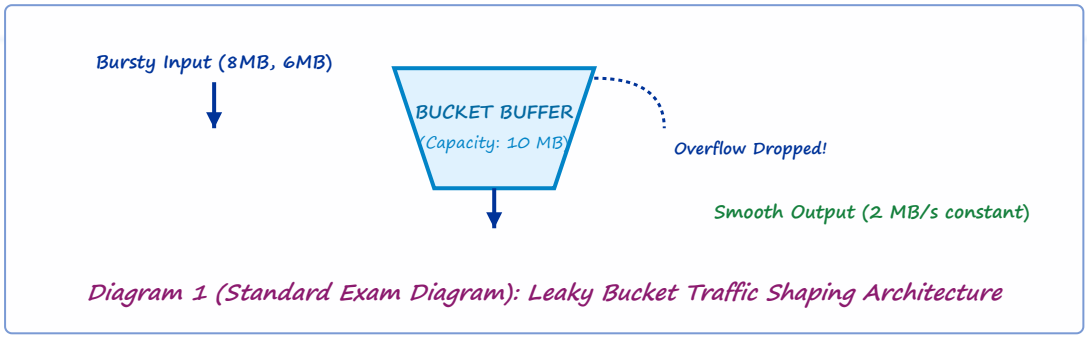


Diagram 1 (Standard Exam Diagram): Leaky Bucket Traffic Shaping Architecture

Diagram 1: Standard Exam Diagram

Shows the physical Leaky Bucket: Bursty input packets entering top, finite bucket capacity buffer, overflow drop path, and constant rate output leak at bottom.

Diagram 2: Easy Understanding Diagram

Shows an irregular funnel turning splashes of water into a smooth, thin trickle.

5. Comparison: Leaky Bucket vs Token Bucket

Feature	Leaky Bucket Algorithm	Token Bucket Algorithm
Output Rate	Strictly **constant & rigid** (no bursts allowed).	Allows **controlled bursts** if tokens are saved.
Discard Policy	Drops packets if bucket is full.	Discards tokens, but holds packets if buffer exists.
Token Mechanism	No tokens; operates purely on water volume.	Generates periodic tokens at a fixed rate.

6. Conclusion

The Leaky Bucket algorithm transforms wildly unpredictable network spikes into a smooth, uniform stream of data, protecting sensitive downstream routers from congestion.

Q4. *(Paper-1)*

Explain any two Approaches which are followed for Congestion control in a computer network?

1. Introduction

When traffic offered to a network exceeds the handling capacity of its communication lines, packet delays soar and throughput drops toward zero. To keep networks running smoothly, network engineers follow two main strategic approaches: **Open-Loop** and **Closed-Loop** Congestion Control.

2. Approach 1: Open-Loop Congestion Control (Prevention)

- **Strategy:** Solve the problem before it can ever start. The system sets conservative rules at connection setup so congestion is virtually impossible.
- **Techniques Used:**
- **Retransmission Policy:** Using conservative timers so false retransmissions do not add extra traffic.
- **Window Policy:** Limiting sender sliding window size (Selective Repeat instead of Go-Back-N).
- **Discarding Policy:** Routers discard less important packets (like video frames) while preserving critical routing updates.
- **Traffic Shaping (Leaky Bucket):** Smoothing user traffic at network edge.

3. Approach 2: Closed-Loop Congestion Control (Detection & Reaction)

- **Strategy:** Actively monitor the network during runtime. When a queue starts backing up, the router triggers feedback signals to throttle senders.
- **Techniques Used:**
- **Choke Packets:** When router queue exceeds a threshold, it sends a high-priority "Choke Packet" directly back to the sender: *"Slow down by 50% immediately!"*
- **Hop-by-Hop Choke:** Each intermediate router slows down the preceding router step-by-step.
- **Explicit Congestion Notification (ECN):** Routers mark two bits in the IPv4/IPv6 header: `ECN = 11` (Congestion Experienced). The receiver echoes this back to the sender in an ACK.

4. Dual Diagrams

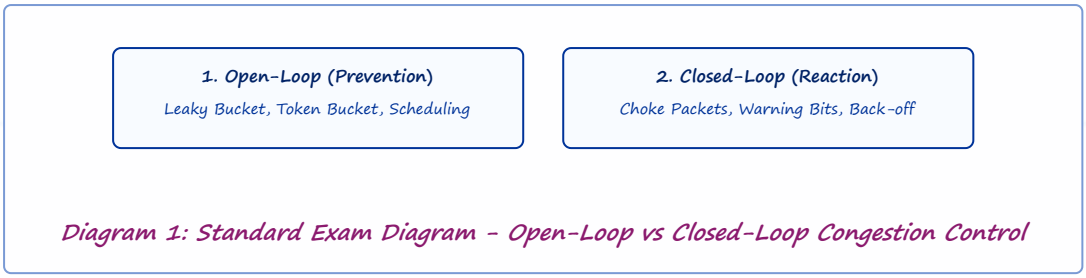


Diagram 1: Standard Exam Diagram - Open-Loop vs Closed-Loop Congestion Control

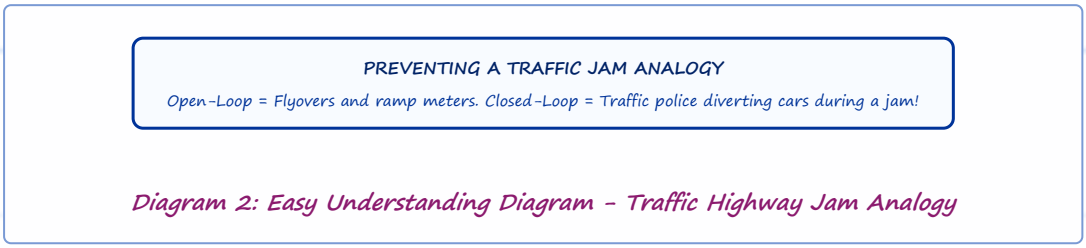


Diagram 2: Easy Understanding Diagram - Traffic Highway Jam Analogy

Diagram 1: Standard Exam Diagram

Shows Open-Loop preventive traffic shaper at the entrance vs Closed-Loop feedback Choke Packet returning from router to sender.

Diagram 2: Easy Understanding Diagram

Shows building fire prevention code (Open-Loop) vs smoke detectors triggering sprinkler alarms (Closed-Loop).

5. Comparison Box: Open-Loop vs Closed-Loop

Feature	Open-Loop Control	Closed-Loop Control
Philosophy	**Prevention:** Stop congestion before it starts.	**Reaction:** Fix congestion after detection.
Runtime Feedback	None.	**Active Feedback:** Choke packets, ECN.
Handling Dynamic Surges	Weak (cannot react to unexpected spikes).	**Strong** (throttles senders dynamically).

6. Conclusion

Modern high-speed networks combine both approaches: Open-loop traffic shaping polices traffic at the network edge, while closed-loop feedback provides real-time protection against internal core router jams.

Q5. *(Paper-2)*

Explain Distance Vector Routing in a computer network with Example?

1. Introduction

Distance Vector Routing (DVR) is a dynamic distributed routing algorithm. Originally developed by Bellman and Ford, it was used in early ARPANET and is the foundation of the Internet's **Routing Information Protocol (RIP)**.

2. Definition & Core Working Principle

- **Principle:** Every router maintains a routing table (vector) containing the shortest known distance to every destination in the network and the next-hop router to take.
- **The Golden Rule:** *"Share your entire routing table with your immediate neighbors only, at periodic intervals (every 30 seconds)."*
- **Bellman-Ford Formula:**

If Router X learns that Neighbor V has a path to Destination Y of cost D_v(Y), and the direct link cost from X to V is c(X, V), the updated distance is:

D_x(Y) = \min_V \{ c(X, V) + D_v(Y) \}

3. Step-by-Step Example

Consider 3 Routers: A, B, C:

- Direct link (A, B) = 2
- Direct link (B, C) = 1
- Direct link (A, C) = 5
- **Initial Table at Router A:** Distance to B = 2, Distance to C = 5 (Direct).
- **Router B shares its table with A:** Router B advertises that its distance to C is **1**.
- **Router A calculates:**

ext{Path via B} = ext{cost}(A, B) + ext{cost}(B, C) = 2 + 1 = \mathbf{3}

- Since 3 < 5, Router A updates its table: **Distance to C = 3 (Next-hop: B)**!

4. Dual Diagrams



Diagram 1: Standard Exam Diagram - Distance Vector Exchange Between Neighbors

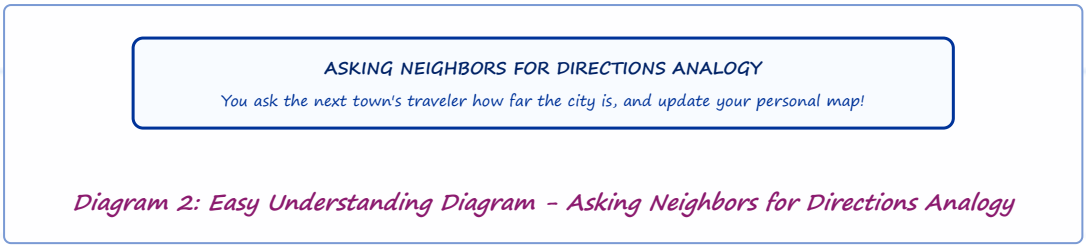


Diagram 2: Easy Understanding Diagram – Asking Neighbors for Directions Analogy

Diagram 1: Standard Exam Diagram

Shows the 3-router triangle network and the updated routing table at Router A showing the shortcut path through B.

Diagram 2: Easy Understanding Diagram

Shows asking a friend for directions: *"I thought the store was 5 km away, but my friend says he is 1 km from the store and 2 km from me, so going through his house is only 3 km!"*

5. Advantages and Disadvantages

Advantages	Disadvantages
Simple Implementation: Easy to configure; low CPU memory extra header data.	**Slow Convergence:** Takes time for updates to ripple across large networks.
Decentralized: Routers operate independently based on neighbor tables.	**Count-to-Infinity Problem:** Link failures can trigger routing loops.

6. Conclusion

Distance Vector Routing offers simple, decentralized routing for small to medium networks, though large networks generally prefer Link State protocols (OSPF) to avoid convergence delays.

Q6. *(Paper-2)*

Explain Token bucket algorithm?

1. Introduction

While the Leaky Bucket algorithm eliminates bursts, it can be too rigid — even if the network is completely empty, it refuses to let a computer send a fast burst. The **Token Bucket Algorithm** improves on this by allowing **controlled bursts of data**, provided the sender has saved up permission tokens.

2. Definition & Working Mechanism

- **Token Bucket:** An algorithm where a bucket holds electronic permission tokens.
- **Token Generation:** Tokens are added to the bucket at a **constant rate of R tokens per second**.
- **Bucket Capacity (B):** The bucket can hold up to a maximum of B tokens. If the bucket is full, newly arriving tokens overflow and are discarded.
- **Packet Transmission Rule:**
- To transmit a packet of size k bytes, the sender must remove k tokens from the bucket.
- If enough tokens exist → The packet is transmitted immediately at full line speed (Burst permitted!).
- If not enough tokens exist → The packet must wait until new tokens arrive.

3. Real-Life Analogy

Think of an **Amusement Park Ride Ticket System**:

- The park gives you 1 ticket every 10 minutes.
- You can save up to a maximum of 5 tickets in your pocket.
- If you sit on a bench and save 5 tickets, you can get on 5 roller coaster rides in a row without waiting (A Burst!).
- Once your tickets are spent, you must wait 10 minutes for your next single ride.

4. Dual Diagrams

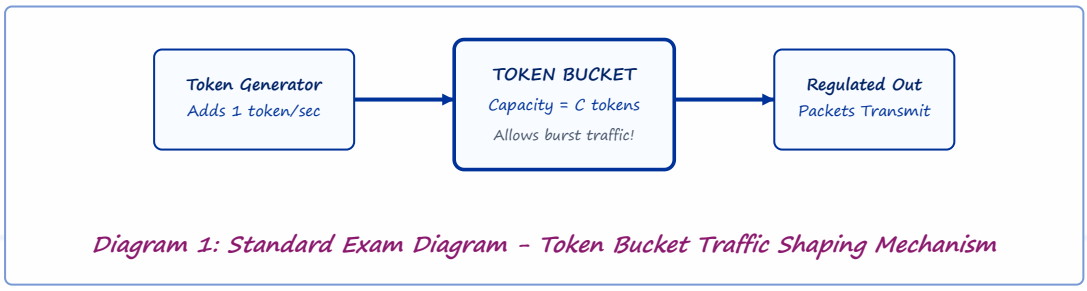


Diagram 1: Standard Exam Diagram – Token Bucket Traffic Shaping Mechanism



Diagram 2: Easy Understanding Diagram – Saving Bus Tokens Analogy

Diagram 1: Standard Exam Diagram

Shows Token Generator adding tokens at rate R, bucket holding tokens up to capacity B, and incoming packet taking tokens before exiting onto the transmission link.

Diagram 2: Easy Understanding Diagram

Shows coin-operated arcade machine: inserting 3 saved tokens lets you play 3 games in a burst.

5. Comparison: Leaky Bucket vs Token Bucket

Feature	Leaky Bucket Algorithm	Token Bucket Algorithm
Discards	Discards **Packets** when full.	Discards **Tokens** when full (never drops user packets).
Bursts Allowed	**Zero bursts allowed** (rigid constant output).	**Allows controlled bursts** up to bucket capacity B.
Output Speed	Fixed, constant leak rate.	Variable speed up to maximum line rate.

6. Conclusion

Token Bucket is the premier traffic shaping and policing algorithm used across the Internet and Quality of Service (QoS) frameworks because it accommodates realistic bursty application traffic without causing packet drops.

Q7. *(Paper-2)*

How is tunneling used in connecting two different networks that speak different protocols (different networks that cannot talk directly)?

1. Introduction

As the Internet evolves, networks with different protocols must connect. For example, modern organizations use **IPv6**, but the core Internet service providers in many regions still run on older **IPv4** routers. **Tunneling** solves this incompatibility without requiring the entire global Internet to be rewired.

2. Definition & Working Mechanism

- **Tunneling:** A technique where a packet formatted for Protocol A is encapsulated inside the payload of a packet formatted for Protocol B so that it can pass through an intermediate network that only understands Protocol B.
- **Three Steps:**
  - **Encapsulation (Entry Gateway):** When an IPv6 packet reaches the boundary router of an IPv4 network, the router slaps a standard 20-byte IPv4 header onto the front of it.

• **Tunneling (Transit):** The intermediate IPv4 routers read only the outer IPv4 header and forward it like any normal IPv4 packet.

• **Decapsulation (Exit Gateway):** At the other end, the exit gateway strips off the outer IPv4 header, recovering the clean original IPv6 packet, and delivers it to the destination IPv6 network.

3. Real-Life Analogy

Think of a **Car on a Ferry Boat**:

- Your car cannot drive across deep ocean water.
- You drive your car onto a ferry ship (**Encapsulation**).
- The ship sails across the ocean (**Tunneling**).
- On the other shore, you drive your car off the ferry onto the road (**Decapsulation**).

4. Dual Diagrams

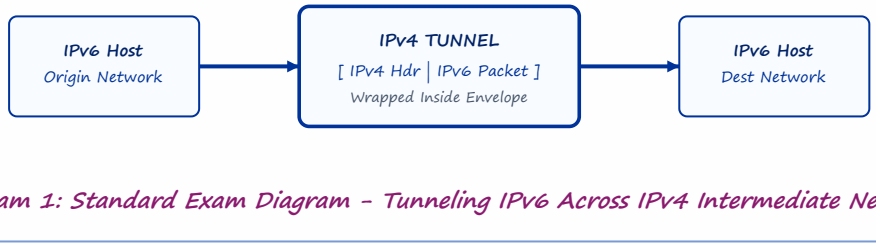


Diagram 1: Standard Exam Diagram – Tunneling IPv6 Across IPv4 Intermediate Network



Diagram 2: Easy Understanding Diagram – Car on a Ferry Boat Analogy

Diagram 1: Standard Exam Diagram

Shows IPv6 Network A → Dual-Stack Router (adds IPv4 header) → IPv4 Internet Tunnel → Dual-Stack Router (strips IPv4 header) → IPv6 Network B.

Diagram 2: Easy Understanding Diagram

Shows an envelope addressed in German placed inside a larger international postal envelope addressed in English.

5. Advantages and Disadvantages of Tunneling

Advantages	Disadvantages
Bridges Incompatible Networks: Allows new protocols (IPv6) to deploy gradually.	**Header Overhead:** Adding an extra outer header reduces usable payload space.
Zero Core Upgrades Needed: Core routers do not need software updates.	Extra processing delay at entry and exit gateway routers.

6. Conclusion

Tunneling is the indispensable bridge of the modern Internet, enabling seamless protocol transitions (IPv6-over-IPv4) and secure private communications (VPNs).

Q8. *(Paper-2)*

Describe the structure of an IPv4 datagram header?

1. Introduction

The **Internet Protocol version 4 (IPv4)** is the foundational protocol of the Internet Network Layer. Every piece of user data traveling across the Internet is packaged inside an **IPv4 Datagram**.

2. Header Size

- **Minimum Header Size: 20 Bytes** (when no optional fields are attached).
- **Maximum Header Size: 60 Bytes** (with up to 40 bytes of options).

3. Detailed Explanation of Header Fields (32-Bit Rows)

Row 1: Format & Length

- **Version (VER — 4 bits):** Stores IP version number (always 4 for IPv4).
- **Header Length (IHL — 4 bits):** Length of header in 4-byte words. Value of 5 means 5 imes 4 = \mathbf{20 ext{ Bytes}}.
- **Type of Service (ToS — 8 bits):** Quality of service flags for priority, delay, and throughput.
- **Total Length (16 bits):** Total size of header + payload in bytes (Maximum size = 65,535 bytes).

Row 2: Fragmentation Controls

- **Identification (16 bits):** Unique ID assigned by sender so fragmented pieces can be recognized.
- **Flags (3 bits):**

- Bit 0: Reserved.
- Bit 1: **DF (Don't Fragment)** — 1 means router cannot fragment.
- Bit 2: **MF (More Fragments)** — 1 means more pieces follow; 0 means last piece.
- **Fragment Offset (13 bits):** Shows where this fragment belongs in the original datagram, measured in 8-byte units.

Row 3: Survival & Routing

- **Time to Live (TTL — 8 bits):** Hop counter decremented by 1 at every router. If it hits 0, the packet is destroyed to prevent endless loops.
- **Protocol (8 bits):** Identifies higher-layer protocol (TCP = 6, UDP = 17, ICMP = 1).
- **Header Checksum (16 bits):** 1's complement checksum protecting only the header.

Rows 4 & 5: IP Addresses

- **Source IP Address (32 bits):** IP address of the sending computer.
- **Destination IP Address (32 bits):** IP address of the destination computer.

4. Dual Diagrams

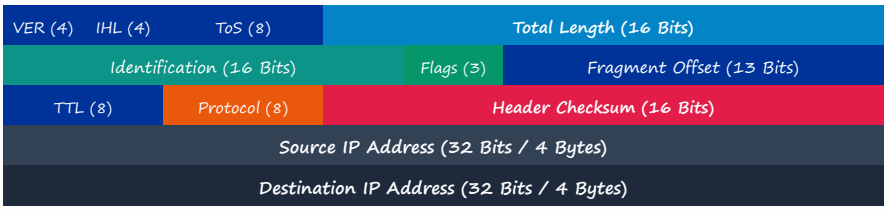


Diagram 1 (Standard Exam Diagram): IPv4 Datagram Header Architecture (Minimum 20 Bytes)

Diagram 1: Standard Exam Diagram

Shows the complete 5-row (20-byte) IPv4 Header architectural grid with all 14 field labels and bit lengths.

Diagram 2: Easy Understanding Diagram

Shows an international postal shipping label: Customs declaration (ToS), Package weight (Total Length), Expiration date (TTL), Contents (Protocol), and Sender/Receiver addresses.

5. Summary Box: IPv4 Header Fields

Field Name	Size (Bits)	Simple Purpose
VER	4	IP version (4).
IHL	4	Header length in 4-byte words (min = 5).
Total Length	16	Header + Data size in bytes.
TTL	8	Hop counter to kill looping packets.
Protocol	8	Transport protocol: TCP (6) or UDP (17).
Source IP	32	Sender address.
Destination IP	32	Receiver address.

6. Conclusion

The IPv4 datagram header contains the essential routing, fragmentation, and survival parameters that enable packets to travel across diverse intermediate routers worldwide.

Q9. *(Paper-3)*

Explain the IPv6 header format in detail.

1. Introduction

By the late 1990s, the explosion of smartphones, laptops, and IoT devices rapidly exhausted IPv4's 4.3 billion address pool. To provide a virtually infinite supply of addresses, the IETF developed **IPv6 (Internet Protocol version 6)**.

2. Key Differences from IPv4

- **Massive Address Space:** 128-bit addresses (3.4 imes 10^{38} addresses — enough to assign billions of IPs to every grain of sand on Earth!).
- **Fixed Header Size:** Fixed strictly at **40 Bytes** (unlike IPv4's variable 20-60 bytes).
- **Simplified Processing:** Only **8 fields** (down from 14 fields in IPv4), enabling hardware routers to route packets at light speed.
- **Zero Checksum:** Checksum was removed because Layer 2 and Layer 4 already check errors.

3. Detailed Explanation of the 8 Fields in IPv6 Header

- **Version (4 bits):** Always set to binary 0110 (representing number 6).
- **Traffic Class (8 bits):** Equivalent to IPv4 Type of Service; used for QoS and priority.

- **Flow Label (20 bits):** Marks a sequence of packets belonging to the same real-time stream (e.g. video call) so routers maintain the same path without recalculating.
- **Payload Length (16 bits):** Size of data payload following the 40-byte header.
- **Next Header (8 bits):** Replaces IPv4's "Protocol" field. Points to the type of extension header or transport protocol (TCP = 6, UDP = 17).
- **Hop Limit (8 bits):** Replaces IPv4's "TTL". Decremented by 1 at each router.
- **Source IP Address (128 bits — 16 Bytes):** Address of the sender.
- **Destination IP Address (128 bits — 16 Bytes):** Address of final recipient.

4. Dual Diagrams

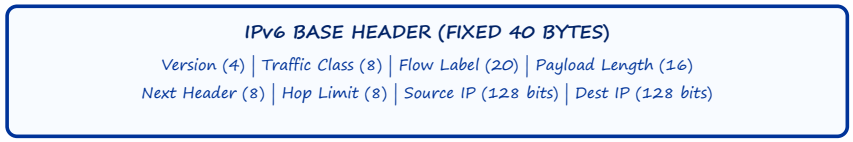


Diagram 1: Standard Exam Diagram - IPv6 Fixed 40-Byte Header Format



Diagram 2: Easy Understanding Diagram - Standard Passport Format Analogy

Diagram 1: Standard Exam Diagram

Shows the streamlined 40-byte IPv6 header grid (Version, Traffic Class, Flow Label, Payload Length, Next Header, Hop Limit, 128-bit Source IP, 128-bit Destination IP).

Diagram 2: Easy Understanding Diagram

Side-by-side visual comparison showing IPv4's cluttered 14-field header versus IPv6's clean, streamlined 8-field layout.

5. Box Comparison: IPv4 Header vs IPv6 Header

Parameter	IPv4 Header	IPv6 Header
Address Size	**32 bits** (4 Bytes) • Approx 4.3 billion IPs	**128 bits** (16 Bytes) • Virtually infinite
Header Size	Variable: **20 to 60 Bytes**.	Fixed: Exactly **40 Bytes**.
Number of Fields	14 fields (cluttered).	**8 fields** (streamlined).

Parameter	IPv4 Header	IPv6 Header
Header Checksum	Included (slows down routers).	**Removed completely** (speeds up routing).
Fragmentation	Done by routers and sender.	Done **only by sender host**.

6. Conclusion

IPv6 solves the global address exhaustion crisis while streamlining router processing through its fixed 40-byte header and removal of checksum extra header data.

Q10. *(Paper-3)*

Explain methods to connect networks.

1. Introduction

A single local area network (LAN) cannot serve an entire university or global corporation. Multiple diverse networks must be interconnected into an internetwork. To achieve this, engineers use specific connecting hardware devices operating at different layers of the OSI model.

2. Four Primary Connecting Devices

A. Repeater / Hub (Layer 1 — Physical Layer)

- **What It Does:** Regenerates electrical or optical signals that have grown weak due to cable attenuation over long distances.
- **Intelligence:** Zero. Just amplifies bits (0s and 1s).

B. Bridge / Switch (Layer 2 — Data Link Layer)

- **What It Does:** Connects two or more LAN segments and filters traffic using hardware **MAC Addresses**.
- **Benefit:** Divides collision domains, preventing packet collisions.

C. Router (Layer 3 — Network Layer)

- **What It Does:** Connects completely different networks (e.g. Ethernet LAN to Internet WAN) and routes packets using **IP Addresses**.
- **Benefit:** Chooses best paths and divides broadcast domains.

D. Gateway (Layer 4 to 7 — Transport to Application Layer)

- **What It Does:** A protocol translator that connects networks running completely incompatible protocol architectures.
- *Example:* Connecting an IBM SNA mainframe network to a TCP/IP network.

3. Dual Diagrams



Diagram 1: Standard Exam Diagram – Distance Vector Exchange Between Neighbors



Diagram 2: Easy Understanding Diagram – Asking Neighbors for Directions Analogy

Diagram 1: Standard Exam Diagram

Shows the 4 devices aligned to their respective OSI layers: Hub/Repeater (Layer 1) • Switch/Bridge (Layer 2) • Router (Layer 3) • Gateway (Layers 4–7).

Diagram 2: Easy Understanding Diagram

Shows small LANs joined by Switches, linked to the Internet via a Router, connecting to a legacy server via a Gateway.

4. Summary Comparison Box: Network Connecting Devices

Device	Operating OSI Layer	Addressing Used	Key Purpose
Repeater / Hub	**Layer 1** (Physical)	None (Raw bits)	Signal regeneration & amplification.
Bridge / Switch	**Layer 2** (Data Link)	48-bit **MAC Address**	Filtering & forwarding within a LAN.
Router	**Layer 3** (Network)	32/128-bit **IP Address**	Best-path routing between networks.
Gateway	**Layers 4 – 7**	Application Profiles	Full protocol translation.

5. Conclusion

From simple signal repeaters to intelligent routers and application gateways, these connecting devices allow thousands of independent local networks to combine into one cohesive global Internet.

Q11. *(MID-2)*

Explain the IPv4 header format and addressing schemes including classful addressing, CIDR, and subnetting.

1. Introduction

Every computer on the Internet must have a unique identifier called an **IP Address**. In IPv4, addresses are 32 bits long, written as four decimal numbers separated by dots (e.g., `192.168.1.10`). Over the years, the way IP addresses are assigned has evolved from **Classful Addressing** to **Subnetting** and **CIDR**.

2. Classful Addressing (The Original System)

In the 1980s, the 32-bit IPv4 address space was divided into 5 rigid classes:

- **Class A (`0.0.0.0` to `127.255.255.255`):**
- Format: `Network.Host.Host.Host` (1st byte Network, 3 bytes Host).
- Supports 128 huge networks with 16 million hosts each.
- **Class B (`128.0.0.0` to `191.255.255.255`):**
- Format: `Network.Network.Host.Host` (2 bytes Network, 2 bytes Host).
- Supports 16,384 medium networks with 65,534 hosts each.
- **Class C (`192.0.0.0` to `223.255.255.255`):**
- Format: `Network.Network.Network.Host` (3 bytes Network, 1 byte Host).
- Supports 2 million small networks with 254 hosts each.
- **Class D (`224.0.0.0` to `239.255.255.255`):** Reserved for **Multicasting**.
- **Class E (`240.0.0.0` to `255.255.255.255`):** Reserved for **Experimental / Research** use.

3. Subnetting (Dividing Networks)

- **Problem with Classful:** If a company needed 300 computers, Class C was too small (max 254) and Class B was absurdly wasteful (65,534 addresses!).
- **Solution:** Subnetting borrows bits from the Host portion to create smaller, manageable subnetworks inside an organization.
- **Subnet Mask:** A bitmask of all 1s for network bits and 0s for host bits (e.g. `255.255.255.0`).

4. CIDR (Classless Inter-Domain Routing — Slash Notation)

- In 1993, CIDR abolished rigid Classes A, B, C entirely.
- Instead of classes, an address is written with a slash suffix (/n) showing how many bits belong to the network prefix:
- 192.168.1.0/24 means the first 24 bits are Network, leaving 32 - 24 = 8 bits for Hosts (2^8 - 2 = 254 hosts).
- 10.0.0.0/8 means the first 8 bits are Network (2^{24} - 2 = 16.7 million hosts).

5. Step-by-Step Subnetting Calculation Example

- **Given Network:** 192.168.10.0/24.
- **Requirement:** Divide into 2 equal subnets.
- **Borrow 1 bit:** Prefix becomes /25 (Subnet mask = 255.255.255.128).
- **Subnet 1:** 192.168.10.0/25 → Usable IPs: 192.168.10.1 to 192.168.10.126.
- **Subnet 2:** 192.168.10.128/25 → Usable IPs: 192.168.10.129 to 192.168.10.254.

6. Dual Diagrams



Diagram 1: Standard Exam Diagram - Distance Vector Exchange Between Neighbors



Diagram 2: Easy Understanding Diagram - Asking Neighbors for Directions Analogy

Diagram 1: Standard Exam Diagram

Shows Class A, B, C bit patterns (Class A begins with 0, Class B with 10, Class C with 110) and CIDR slash notation hierarchy.

Diagram 2: Easy Understanding Diagram

Shows an apartment building floor divided into smaller rooms (Subnets) with room numbers (Host IDs).

7. Box Comparison: Classful vs CIDR

Parameter	Classful Addressing	Classless Addressing (CIDR)
Flexibility	Rigid fixed sizes (A, B, C).	**Arbitrary flexible block sizes** (/1 to /32).
Address Wastage	Enormous (wasted millions of IPs).	Minimal (organizations buy exact sizes).
Notation	Class name + default mask.	Slash notation (e.g., `192.168.1.0/26`).

8. Conclusion

Subnetting and CIDR saved the IPv4 Internet from premature collapse by eliminating wasteful class boundaries and extending IPv4's lifespan until IPv6 adoption.

Q12. *(MID-2)*

Discuss various routing algorithms used in the Network Layer (shortest path, distance vector, and link state).

1. Introduction

Routing is the core job of the Network Layer: finding the most efficient path for packets to travel from source to destination across multiple router hops. Routing algorithms are classified into **Shortest Path**, **Distance Vector**, and **Link State**.

2. Breakdown of the Three Algorithms

A. Shortest Path Routing (Dijkstra's Algorithm)

- **Type:** Centralized / Graph search algorithm.
- **Working:** Builds a tree of shortest paths from source to all other nodes by incrementally adding the closest unvisited node.
- **Strength:** Guaranteed to find the minimum-cost loop-free path.

B. Distance Vector Routing (Bellman-Ford / RIP)

- **Type:** Distributed iterative algorithm.
- **Working:** Routers share their entire routing table with immediate neighbors only, at regular periodic intervals (every 30s).
- **Weakness:** Slow convergence and vulnerable to the **Count-to-Infinity** problem.

C. Link State Routing (Dijkstra / OSPF)

- **Type:** Global distributed algorithm.

- **Working:** Every router discovers its immediate neighbors and floods this link-state information to **every router in the entire network**.
- Every router builds an identical complete map of the entire network topology and runs Dijkstra locally.
- **Strength:** Converges in milliseconds with zero loops!

3. Dual Diagrams



Diagram 1: Standard Exam Diagram – Distance Vector Exchange Between Neighbors



Diagram 2: Easy Understanding Diagram – Asking Neighbors for Directions Analogy

Diagram 1: Standard Exam Diagram

Shows Link State Router flooding LSP packets across the network, contrasted with Distance Vector sharing local vectors between neighbors.

Diagram 2: Easy Understanding Diagram

Shows Distance Vector as asking strangers on the street for directions vs Link State as downloading a complete Google Map of the whole city.

4. Master Comparison Box of Routing Algorithms

Parameter	Shortest Path (Dijkstra)	Distance Vector (RIP)	Link State (OSPF)
Knowledge of Topology	Full network graph.	Knows only neighbor distances.	**Full identical map of entire network.**
Information Shared	Local calculation.	Entire routing table to neighbors.	Link state of neighbors to **all routers.**
Convergence Speed	Immediate.	**Slow** (can form loops).	**Ultra-Fast** (milliseconds).

Parameter	Shortest Path (Dijkstra)	Distance Vector (RIP)	Link State (OSPF)
Traffic Overhead	None (runs locally).	Low on small links, slow on large.	Flooding consumes burst bandwidth.
Industry Protocol	Theoretical base.	**RIP** (Routing Information Protocol).	**OSPF**, IS-IS.

5. Conclusion

While Distance Vector is simple for small local networks, Link State routing (OSPF) dominates modern enterprise backbones because it converges almost instantaneously and eliminates routing loops.